

Theorem (Markov's Inequality). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X : \Omega \rightarrow [0, \infty]$ be a non-negative random variable with finite expectation $\mathbb{E}[X]$. For any constant $a > 0$,*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Equality holds iff $X = a \mathbf{1}_{\{X \geq a\}}$ a.s. (i.e. X takes only the two values 0 and a).

Proof. Assumptions and edge-case analysis. We assume $X(\omega) \geq 0$ for every $\omega \in \Omega$ and $a > 0$. If $\mathbb{P}(X \geq a) = 0$ the inequality is immediate. If $\mathbb{E}[X] = \infty$ the right-hand side is $+\infty$ and the inequality holds trivially. The case $a = 0$ is excluded by hypothesis.

Indicator of the event $\{X \geq a\}$. Define the random variable

$$\mathbf{1}_{\{X \geq a\}}(\omega) = \begin{cases} 1, & \text{if } X(\omega) \geq a, \\ 0, & \text{otherwise.} \end{cases}$$

By the definition of expectation of an indicator,

$$\mathbb{E}[\mathbf{1}_{\{X \geq a\}}] = \mathbb{P}(X \geq a). \quad (2.1)$$

Pointwise domination. For each $\omega \in \Omega$ we have

$$X(\omega) \geq a \mathbf{1}_{\{X \geq a\}}(\omega). \quad (3.1)$$

Verification. If $X(\omega) \geq a$ then $\mathbf{1}_{\{X \geq a\}}(\omega) = 1$ and the right-hand side equals $a \leq X(\omega)$. If $X(\omega) < a$ then the indicator is zero and the inequality reduces to $X(\omega) \geq 0$, which holds by non-negativity of X .

Taking expectations and applying monotonicity. Since (3.1) holds pointwise, monotonicity of expectation yields

$$\mathbb{E}[X] \geq \mathbb{E}[a \mathbf{1}_{\{X \geq a\}}]. \quad (4.1)$$

Linearity of expectation gives

$$\mathbb{E}[a \mathbf{1}_{\{X \geq a\}}] = a \mathbb{E}[\mathbf{1}_{\{X \geq a\}}]. \quad (4.2)$$

Substituting (2.1) into (4.2) and then into (4.1) we obtain

$$\mathbb{E}[X] \geq a \mathbb{P}(X \geq a). \quad (4.3)$$

Division by the positive constant a . Because $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}. \quad (5.1)$$

This is precisely Markov's inequality.

Equality condition. Equality in (5.1) can occur only if every inequality used above is an equality.

Equality in the monotonicity step (4.1) requires that the pointwise inequality (3.1) be an equality $\mathbb{P}$ -almost everywhere, i.e.

$$X(\omega) = a \mathbf{1}_{\{X \geq a\}}(\omega) \quad \text{a.s.} \quad (6.1)$$

Thus X takes only the values 0 and a almost surely. Conversely, if X assumes only the two values 0 and a with probabilities p_0 and p_a ($p_0 + p_a = 1$), then $\mathbb{P}(X \geq a) = p_a$ and $\mathbb{E}[X] = a p_a$, so (5.1) holds with equality.

Hence equality holds *iff*

$$X = a \mathbf{1}_{\{X \geq a\}} \quad \text{a.s.},$$

i.e. the random variable takes only the values 0 and a . $\square$

$\square$